

Plan: 3-Dataset HSIKAN Validation
(Slashdot + Bitcoin{Alpha,OTC} + Epinions)
v1 vertex prefilter $\times$ v2 FPN tiered $\times$ SoA

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1 Scope and goal

Validate the v1 vertex prefilter + v2 FPN tiered caps + SoA cycle enumerator refactor (all landed 2026-05-10..11) against the three datasets where HSiKAN currently holds SOTA (Slashdot, Bitcoin Alpha, Bitcoin OTC) under the transductive protocol, and probe whether the v2 tiered ladder closes any of the documented -0.094 Epinions AUC gap to SGT.

Goals:

1. **AUC parity confirmation:** v1 (vertex prefilter), v2 (FPN tiered), and v1+v2 combined preserve the 5-seed AUC of the baseline configuration within ± 0.005 at the production config on each dataset. The SoA refactor is bit-identical-output by construction (unit-tested 2026-05-11), so parity here is about the cycle-pool shifts induced by v1/v2, not the SoA path.
2. **Wall-time savings:** measure the cumulative wall win from SoA + dedup-stack-key + v1/v2 vs the legacy HSIKAN_TOPK_MODE=per_vertex path. Expected $\sim 21\%$ on Slashdot c4 (already measured); confirm at full 5-seed scale. Expected on Epinions: $\sim 30\text{--}40\%$ enum savings (extrapolated from 22% Slashdot triangle filter at higher graph sparsity).
3. **Epinions gap probe:** test if v2 FPN tiered + Ricci-style shape-prior insights from the HyMeYOLO work lifts Epinions AUC above the 0.8464 baseline. This is a stretch goal — the ceiling-claim memory (project.epinions.ceiling.2026.05.09) says the gap needs sparse-attention or learnable-incidence; v2 by itself may not close it but the data is informative.

2 Configurations

Four configurations per dataset, exact env-var recipes:

```
# B (baseline)      reproduces yesterday's 5-seed SOTA exactly
HSIKAN_TOPK_MODE=per_vertex
HSIKAN_TOPK_K=128
HSIKAN_TOPK_PRUNER=balance
HSIKAN_TOPK_SCORER=fraction_negative
HSIKAN_TRITON_KERNEL=1
HSIKAN_TRITON_BACKWARD=1
# (no HSIKAN_VERTEX_FILTER, no HSIKAN_TOPK_TIERS)

# V1 = baseline + vertex prefilter (compose: degree + triangle)
HSIKAN_VERTEX_FILTER=compose:degree,triangle
HSIKAN_VERTEX_FILTER_MIN_DEGREE=2

# V2 = baseline + FPN-style concentric per-vertex caps
HSIKAN_TOPK_MODE=per_vertex_tiered
HSIKAN_TOPK_TIERS=0.1:1024,1.0:256,5.0:64,20.0:16,100.0:0
```

```
# V1+V2 = both stacked
HSIKAN_TOPK_MODE=per_vertex_tiered
HSIKAN_TOPK_TIERS=0.1:1024,1.0:256,5.0:64,20.0:16,100.0:0
HSIKAN_VERTEX_FILTER=compose:degree,triangle
HSIKAN_VERTEX_FILTER_MIN_DEGREE=2
```

All four configs use the SoA-batched PyO3 path (landed 2026-05-11); the legacy non-batched output path is no longer reachable from the canonical bindings. AUC is invariant to this; wall is not.

3 Datasets

Dataset	$ V $	$ E $	SOTA baseline AUC (5-seed)
Bitcoin Alpha	3,783	24,186	0.9468 (cycle), 0.9845 (joint)
Bitcoin OTC	5,881	35,592	0.9266 (cycle), 0.9801 (joint)
Slashdot	82,140	549,202	0.9070 ± 0.0029 (edge_cr, kernel on)
Epinions	131,828	841,372	0.8464 ± 0.0106 (edge_cr)

4 Affected files

- `signedkan_wip/experiments/run_3dataset_v1_v2_phase_{A,B,C,D}.sh` — NEW orchestrator scripts, one per phase below.
- `signedkan_wip/experiments/results/3dataset_v1_v2_{phase_A,phase_B,phase_C,phase_D}.jsonl` — per-phase result lines (one per (config, dataset, seed) triple).
- `reports/2026-05-13-3dataset-v1-v2.md` — final report with per-dataset paired tables, wall ratios, AUC deltas, and promotion verdicts.
- No source changes required — v1/v2/SoA are already shipped end-to-end as of 2026-05-11 commits.

5 CORE.YAML items touched

None. All affected files are in `signedkan_wip/experiments/` or `reports/` (allowlist per CORE.YAML).

6 Test strategy (CLAUDE.md §3)

Acceptance per phase is gated on paired AUC + wall measurements, not on unit tests (the underlying code is already tested: 117 Rust + 12 cycle_cache + cluttered_mnist + circles_ricci = > 150 tests passing as of 2026-05-11).

Within each script:

- Seed-0 smoke fires before the multi-seed loop; if seed-0 fails to converge (loss does not decrease over the first 10 epochs), abort the script and dump the log path.
- Per-seed JSON line written immediately on completion so a partial crash doesn't lose results.
- Final aggregate (mean $\pm$ stdev) computed and emitted at script end; comparison to recorded baselines automatic.

7 Ladder

Phase A — 1-seed smoke across all 4 datasets

Goal: confirm all 4 configs train end-to-end on all 4 datasets at production config; surface any config $\times$ dataset combinations that drop > 0.005 AUC vs baseline single-seed.

Dataset	Per-config wall	4 configs total
Bitcoin Alpha	~ 1 min	~ 4 min
Bitcoin OTC	~ 2 min	~ 8 min
Slashdot	~ 13 min	~ 52 min
Epinions	~ 115 min	~ 460 min (~ 7.7 h)
Total		~ 9 h

Acceptance gate: every (config, dataset) pair completes without OOM; AUC delta within ± 0.01 of baseline single-seed. If a single config OOMs or crashes on a dataset, that config is flagged and excluded from later phases on that dataset.

Phase B — 5-seed paired on BitcoinAlpha,OTC

Goal: paired statistical confirmation that V1, V2, V1+V2 preserve AUC at 5σ -style paired tightness on the cheap datasets where wall is no constraint.

Dataset	5-seed $\times$ 4 configs	Total
Bitcoin Alpha	5 min	20 min
Bitcoin OTC	10 min	40 min
Total		~ 1 h

Acceptance gate: paired delta vs B (baseline) within ± 0.005 for V1, V2, V1+V2 at 5-seed; published SOTA (0.9845 / 0.9801 joint cycle+walk) preserved. If V1+V2 fails parity, halt Phase C until investigated.

Phase C — 5-seed paired on Slashdot

Goal: the medium-sized dataset where v1/v2 already showed wall wins. Paired AUC parity + wall measurement.

5-seed $\times$ 4 configs	Total wall
~ 11 min / seed average $\times$ 20 seeds	~ 3.7 h

Acceptance gate:

- V1+V2 wall $\leq 0.80 \times$ baseline wall.
- V1+V2 AUC paired delta within ± 0.002 of 0.9070 baseline ($\sigma_{\text{baseline}} = 0.0029$, so ± 0.002 is roughly half a sigma).
- If V1+V2 lifts AUC by ≥ 0.003 paired: **paper headline** candidate.
- If V1+V2 drops AUC by > 0.005 : V1 alone may still be the right shipping config; document and proceed to Phase D with V1 only.

Phase D — Epinions 5-seed (2 configs only)

Goal: the gap-closing experiment.

Pre-cache (obsolete): 5 seeds $\times$ 2 configs $\times$ 115 min/seed $= \sim 19$ h. **Post-cycle_cache wiring** (2026-05-11): seed-0 enumerates (~ 95 min on Epinions), seeds 1-4 hit the cache and skip directly to training (~ 20 min each).

Two-config compromise: B (baseline, reproduces 0.8464) vs the best of {V1+V2, V1}-from-Phase-A. 10 seeds total.

5-seed $\times$ 2 configs (post-cache)	Total wall
2 configs $\times$ (95 + 4 $\times$ 20) min	~ 6.5 h

Acceptance gate:

- V1+V2 lifts AUC by ≥ 0.010 paired vs baseline 0.8464 ($\sigma_{\text{baseline}} = 0.0106$, so ≥ 0.010 is paired $\sigma \approx 2$): partial gap closure, publish.
- OR V1+V2 wall $\leq 0.70\times$ baseline wall at AUC parity: practical engineering win, publish as a methods contribution.
- Both: ideal outcome, paper headline.
- Neither: document as null, preserves the `project_epinions_ceiling_2026-05-09` ceiling claim — gap requires sparse-attention or learnable-incidence, not cycle-pool refinement.

8 Performance budget

Update 2026-05-11 post-cycle_cache wiring: with `HYMEKO_CYCLE_CACHE=1` now wired into `run_final_cell.py` (`cell_signed_graph` + `cell_scene` call sites), multi-seed runs amortise enumeration over seeds 2–5. Cache hit verified end-to-end: seed-0 misses (writes `.npz`), seed-1+ hit (loads 720 KB instead of re-enumerating). For Epinions this drops 5-seed wall from $5 \times 115 = 575$ min to $115 + 4 \times 20 = 195$ min ($\sim 66\%$ reduction).

Budget table (post-cache):

- Phase A (1-seed, no cache benefit): ≤ 10 h wall total.
- Phase B (5-seed Bitcoin, 4 configs): ~ 0.5 h.
- Phase C (5-seed Slashdot, 4 configs): ~ 1.5 h.
- Phase D (5-seed Epinions, 2 configs): **~ 6.5 h.**
- Peak GPU memory: ≤ 6 GB (RTX 2070 SUPER consumer-grade).
- Peak system RSS: ≤ 12 GB (the 2026-05-10 morning OOM at 22 GB must not recur; `ulimit -v 16777216` on every script).

Cumulative total: **~ 17 h (post-cache)** vs the pre-cache projection of ~ 33 h. Phase D alone drops from 19 h to 6.5 h.

Caveat: cache invariant is that `(graph_hash, k, cap, scorer, pruner, topk_mode, ...)` is identical across seeds. Switching configs (B $\rightarrow$ V1 $\rightarrow$ V2 $\rightarrow$ V1+V2) intentionally creates a different cache key per config; each config still does its first-seed enumeration. This is the desired behaviour for paired statistical comparison.

9 Risk anticipation (CLAUDE.md §2)

Performance-contract preservation

The 2026-05-10 Python-side `_subsample_arr_to_tuples` landed; the 2026-05-11 SoA refactor + dedup-stack-key landed. **Both contracts must be honoured in the run scripts:** do not set `HSIKAN_TOPK_K` above 128 unless explicitly authorised (Rust-side per-vertex heaps grow $O(K \cdot |V|)$).

Aspirational comments to honour

`project_rust_topk_memory_followup_2026-05-10`: the Rust-side `cap_total` pushdown has NOT been implemented. Peak Rust RSS during $k=5$ enumeration on Epinions can hit 6–10 GB even with the Python-side fix. Phase A’s Epinions runs are the first end-to-end test of whether this is OK at production config with v1/v2 on; if Phase A Epinions OOMs, Rust-side cap pushdown becomes a Phase D prerequisite.

Worst-case input

Epinions production-config c2-c5 + w2,w3 mixed-arity training, 80 epochs, max_k4=100k, quaternion attention + edge_cr highway, Triton kernel on. V2's tiered caps will shift the cycle pool significantly on $k=3/k=5$ (lift study probe data) but is untested against the actual training loop at production scale.

What could go wrong

- **V2 tiered changes cycle quality:** the tier thresholds (0.1%, 1%, 5%, 20%, 100%) are taken from the lift-studies report's predicted-AUC sweet spot but never AUC- measured at production. V2 single-seed Phase A is the binary go/no-go.
- **V1 vertex prefilter drops too much:** the `compose:degree,triangle` default may be too aggressive on Epinions's sparse structure (Epinions has more leaves and bridge-like vertices than Slashdot). Adaptive fallback: `min_degree = 1` if Phase A flags it.
- **The Triton kernel + tiered + filter combination has not been tested together at production scale.** Phase A is the first integration test.

10 Empty-plan-dir hygiene

This plan dir is fully populated. If abandoned at any halt condition, the dir gets a `HALTED.md` explaining the abort.

11 Halt conditions

- Any Phase-A config OOMs on any dataset $\rightarrow$ halt that (config, dataset) pair; document and proceed without it.
- Bitcoin 5-seed (Phase B) shows V1+V2 dropping > 0.01 paired AUC $\rightarrow$ halt before Phase C; design fix required.
- Slashdot 5-seed (Phase C) shows V1+V2 dropping > 0.005 paired AUC $\rightarrow$ pick the best non-V2 config (V1 alone or B) for Phase D; do not run V2 on Epinions.
- Epinions Phase D individual seed wall projection > 4 h $\rightarrow$ cancel, fall back to 1-seed each at Epinions.

12 Rollback path

No code changes ship with this plan; only experiment scripts and the final report. Revert is `rm -rf docs/plans/2026-05-12-3dataset-hsikan-v1-v2/` + corresponding experiment-script removals.

13 Implementation order

1. Write Phase A orchestrator script and one-shot test the smallest dataset (Bitcoin Alpha, 1-seed) to confirm the recipe works. ~ 5 min.
2. Phase A full sweep (~ 9 h, overnight).
3. Decision gate: review Phase A AUC deltas, identify any config $\times$ dataset failures, lock the configs for Phase B–D.
4. Phase B (1 h).
5. Phase C (4 h).
6. Phase D (19 h, requires explicit user authorisation for the long Epinions run — not auto-launched).

7. Final report at `reports/2026-05-13-3dataset-v1-v2.md`.

All phases gated on user authorisation for the longer runs (CLAUDE.md §11 halt: “unverified experiment that mutates persistent state”).